IGCSE 0607 Practice Questions — E2.1 — Combined Questions & Markscheme

Introduction to algebra

Total Questions: 30 — **Total Marks:** 131 Marks

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- **E2.1** Introduction to algebra
  → **Paper 2 (Non-Calculator)** *(18 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

1 Work out the value of the expression

$$3x^2 - 5xy$$

when $x = -4$ and $y = 2$.

..... **[2]**

2 Work out the value of

$$5x^2 - 3y$$

when $x = -2$ and $y = 4$.

..... **[2]**

3 Find the value of $3a - 2b$ when $a = 4$ and $b = -5$.

..... **[2]**

4 Work out the value of

$$3x^2 - 4y$$

when $x = -3$ and $y = -5$.

..... **[2]**

5 Work out the value of

$$\frac{3x - y^2}{2z}$$

when $x = 5$, $y = -4$ and $z = 0.5$.

..... **[2]**

6 Given the formula $v = u + at$.

Find the value of v when $u = 5$, $a = -2$, and $t = 3$.

$v =$ ..... **[3]**

7 **(a)** Work out the value of $4x - 3y$ when $x = -2$ and $y = 5$.

..... **[2]**

(b) Work out the value of $\dfrac{p^2 - 7}{2q}$ when $p = 5$ and $q = -3$.

..... **[2]**

8 **(a)** Work out the value of $3a - 5b$ when $a = -2$ and $b = 4$.

..... **[2]**

(b) Work out the value of $x^2 + 4x$ when $x = -3$.

..... **[2]**

9 Consider the formula

$$P = 3a^2 - 2ab + 5b$$

(a) Calculate the value of P when $a = 4$ and $b = -3$.

$P =$ ..... **[2]**

(b) Calculate the value of b when $P = 19$ and $a = -2$.

$b =$ ..... **[2]**

10 **(a)** Work out the value of $5x - 2y$ when $x = 3$ and $y = -4$.

..... **[2]**

(b) Evaluate

$$\frac{a^2 - b^3}{2c}$$

when $a = -4$, $b = 2$ and $c = 3$.

..... **[2]**

11 (a) (a) The cost of hiring a plumber is calculated using the formula

$C = 40 + 25h$ where C is the cost in dollars and h is the number of hours worked.
Work out the cost of hiring a plumber for 3.5 hours.

$C = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

(b) Given that $v = u + at$, find the value of v when $u = -12$, $a = 9.8$ and $t = 3$.

$v = \ldots\ldots\ldots\ldots\ldots$ **[3]**

12 **(a)** A plumber charges a fixed fee of \$40 plus \$25 for each hour he works.
Write down an expression, in terms of h, for the total cost, in dollars, for h hours of work.

\$ ..... **[1]**

(b) Given the formula

$$P = \frac{k^3 - m^2}{\sqrt{n}}$$

Work out the exact value of P when $k = -2$, $m = -3$ and $n = \frac{1}{16}$.

$P =$ ..... **[3]**

13 Find the exact value of

$$W = \frac{p^{-2} + q^3}{\sqrt[3]{r}}$$

when $p = \frac{1}{2}$, $q = -2$ and $r = -\frac{1}{8}$.

$W =$ ..... **[4]**

14 The distance, d, between two points (x_1, y_1) and (x_2, y_2) is given by the formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(a) Work out the value of d when $x_1 = 4$, $y_1 = -3$, $x_2 = -8$, and $y_2 = 2$.

$d =$ ..... **[2]**

(b) Given that $d = 17$, $x_1 = k$, $y_1 = 8$, $x_2 = 3$, and $y_2 = -7$, work out the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

15 The nth term of a sequence is given by the expression

$$T_n = an^2 + bn + c$$

where a, b, and c are constants.
The first three terms of the sequence are $T_1 = 4$, $T_2 = 13$, and $T_3 = 28$.

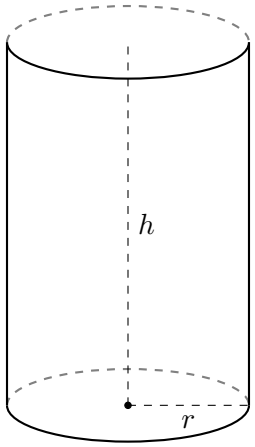
(a) By substituting $n = 1$, $n = 2$, and $n = 3$ into the expression, find the values of a, b, and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

(b) Find the value of k such that $T_{k+1} - T_k = 57$.

$k =$ ..... **[2]**

16 A cylindrical container of radius r and height h is open at the top.
The total exterior surface area of the container, A, consists of the curved surface area and the circular base.



(a) Show that the total surface area A can be written as

$$A = \pi r(2h + r)$$

[2]

(b) **(i)** Calculate the exact surface area A when $r = 4$ and $h = 6$.
Give your answer in terms of π.

$A =$ ..... **[2]**

(ii) The volume of the cylinder is V.
The formula for the volume is $V = \pi r^2 h$.
Express h in terms of V, π and r, and substitute this into the expression for A to show that

$$A = \frac{2V}{r} + \pi r^2$$

[3]

17 Consider the expression $P = \dfrac{2x^2 - 3y}{x - y}$.

(a) Calculate the value of P when $x = -2$ and $y = 4$.

$P =$ ..... **[3]**

(b) Rearrange the formula to make y the subject.

$y =$ ..... **[4]**

18 The mathematical operation $*$ is defined for all non-zero numbers a and b as

$$a * b = \frac{a^2 - b^2}{ab}$$

(a) Evaluate the following, giving each answer as a fraction in its simplest form.

(i) $6 * (-2)$

..... **[2]**

(ii) $\frac{1}{2} * 4$

..... **[3]**

(b) Given that $x * 2 = \frac{3}{2}$, find the positive value of x.

$x =$ ..... **[3]**

19 A plumber charges a fixed call-out fee of $\$C$ and an hourly rate of $\$R$.

(a) Write down an expression for the total cost, T dollars, for a job that takes h hours.

$T =$ ..... **[1]**

(b) The call-out fee is \$45 and the hourly rate is \$32.50.
Work out the total cost for a job that takes 3.5 hours.

\$ ..... **[2]**

20 A formula is given by

$$P = \frac{\sqrt{x - y}}{z^3}$$

Calculate the value of P when $x = 12.5$, $y = 3.84$, and $z = -1.2$.
Give your answer correct to 3 significant figures.

$P =$ ..... **[3]**

21 A solid shape has a volume, V, given by the formula

$$V = \pi r^2 h + \frac{2}{3}\pi r^3$$

Work out the value of V when $r = 4.5$ and $h = 8.2$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

22 **(a)** A plumber charges a fixed fee of \$45 plus \$32 for each hour worked.
Write down a formula for the total charge, C dollars, when the plumber works for h hours.

$C =$ ..... **[1]**

(b) Work out the total charge when the plumber works for 3.5 hours.

\$ ..... **[1]**

(c) Work out the value of

$$x^3 - 4xy$$

when $x = -3$ and $y = 2.5$.

..... **[2]**

23 The volume, V, of a cone with radius r and height h is given by the formula

$$V = \frac{1}{3}\pi r^2 h$$

(a) Calculate the value of V when $r = 6.4$ and $h = 12.6$.
Give your answer correct to 1 decimal place.

$V =$ ..... **[2]**

(b) Work out the value of r when $V = 800$ and $h = 14$.
Give your answer correct to 3 significant figures.

$r =$ ..... **[3]**

24 The cost, C dollars, of hiring a campervan for d days and travelling m kilometres is given by the formula

$$C = 85d + 0.24m$$

(a) Calculate the cost of hiring the campervan for 7 days and travelling 450 kilometres.

$ ..... **[2]**

(b) A family hires the campervan for 14 days.
The total cost is $1475.60.
Work out the number of kilometres they travelled.

..... km **[3]**

25 The cost, C dollars, of hiring a plumber for h hours is given by the formula:

$C = 45 + 35h$

(a) Calculate the cost of hiring the plumber for 3.5 hours.

$C =$ ..... **[2]**

(b) The plumber charges a total of \$325 for a job.
Calculate the number of hours the plumber worked.

$h =$ ..... hours **[3]**

26 The total surface area, A cm^2, of a solid cone with base radius r cm and vertical height h cm is given by the formula

$$A = \pi r \left(r + \sqrt{r^2 + h^2} \right)$$

(a) Calculate the value of A when $r = 5$ and $h = 12$.
Give your answer correct to 3 significant figures.

$A =$ ..... **[2]**

(b) Another solid cone has base radius $r = 6$ and total surface area $A = 96\pi$.
Calculate the vertical height, h, of this cone.

$h =$ ..... **[4]**

27 The volume, V, of a spherical cap of height h and base radius a is given by the formula

$$V = \frac{1}{6}\pi h(3a^2 + h^2)$$

(a) Calculate the volume of a spherical cap with height $h = 4.2$ cm and base radius $a = 6.5$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

(b) Rearrange the formula to make a^2 the subject.

$a^2 =$ ..... **[3]**

(c) Find the base radius, a, of a spherical cap with volume $V = 150$ cm^3 and height $h = 3$ cm. Give your answer correct to 3 significant figures.

$a =$ ..... cm **[1]**

28 The value of y is given by the formula

$$y = \frac{a - b^2}{c - \sqrt{d}}$$

Work out the value of y when $a = -4.5$, $b = -3.2$, $c = 1.8$ and $d = 0.64$.

$y =$ ..... **[3]**

29 A formula used in physics is

$$s = ut + \frac{1}{2}at^2$$

(a) **(a)** Calculate the value of s when $u = 5.2$, $t = 4.5$ and $a = -9.8$.

$s =$ ..... **[2]**

(b) Calculate the positive value of t when $s = -9.8$, $u = 4.9$ and $a = -9.8$.

$t =$ ..... **[3]**

30 The volume, V, and surface area, A, of a solid are given by the formulas below.

(a) The formula for the volume is

$$V = \frac{1}{3}\pi h(r^2 + rR + R^2)$$

Calculate the value of V when $h = 12.5$, $r = 4.2$, and $R = 7.8$.
Give your answer correct to 3 significant figures.

$V =$ ..... **[3]**

(b) The formula for the surface area is

$$A = \pi(R + r)\sqrt{(R - r)^2 + h^2} + \pi r^2 + \pi R^2$$

Calculate the value of A when $h = 12.5$, $r = 4.2$, and $R = 7.8$.
Give your answer correct to 1 decimal place.

$A =$ ..... **[3]**

1 M1 for $3(-4)^2$ or $-5(-4)(2)$ or 48 or $+40$

88 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Correct substitution: $5(-2)^2 - 3(4)$ or $5(4) - 12$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 2]*

3 $3(4) - 2(-5)$ or $12 - -10$ or $12 + 10$ **[M1]**

22 **[A1]**

[Back to Question] *[Total Marks: 2]*

4 Substitutes values correctly, e.g. $3(-3)^2 - 4(-5)$ **[M1]**

47 **[A1]**

[Back to Question] *[Total Marks: 2]*

5 Substitution of all three values correctly e.g. $\dfrac{3(5) - (-4)^2}{2(0.5)}$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 2]*

6 Substitute the given values into the formula: $v = 5 + (-2)(3)$ **[M1]**

$v = 5 - 6$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 3]*

7 (a) $4(-2) - 3(5)$ or $-8 - 15$ **[M1]**

-23 **[A1]**

(b) $\dfrac{5^2 - 7}{2(-3)}$ or $\dfrac{25 - 7}{-6}$ **[M1]**

-3 **[A1]**

[Back to Question] *[Total Marks: 4]*

8 (a) -6 or -20 seen **[M1]**

-26 **[A1]**

(b) $(-3)^2 = 9$ or $4(-3) = -12$ seen **[M1]**

-3 **[A1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** $P = 3(4)^2 - 2(4)(-3) + 5(-3)$ or $48 + 24 - 15$ seen **[M1]**

57 **[A1]**

(b) $19 = 3(-2)^2 - 2(-2)b + 5b$ or $19 = 12 + 9b$ **[M1]**

$\frac{7}{9}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** Substitutes values correctly: $5(3) - 2(-4)$ **[M1]**

$15 + 8 = 23$ **[A1]**

(b) Substitutes values: $\frac{(-4)^2 - (2)^3}{2(3)}$ **[M1]**

Evaluates numerator: $16 - 8 = 8$

$\frac{8}{6} = \frac{4}{3}$ or $1\frac{1}{3}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** **(a)** $40 + 25(3.5)$ or $40 + 87.5$ **[M1]**

127.50 **[A1]**

(b) $-12 + 9.8 \times 3$ **[M1]**

$-12 + 29.4$ **[M1]**

17.4 **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** $40 + 25h$ **[B1]**

(b) $(-2)^3 = -8$ and $(-3)^2 = 9$ **[M1]**

$\sqrt{\frac{1}{16}} = \frac{1}{4}$ **[B1]**

-68 **[A1]**

[Back to Question] *[Total Marks: 4]*

13 Evaluates $p^{-2} = \left(\frac{1}{2}\right)^{-2} = 4$ **[B1]**

Evaluates $q^3 = (-2)^3 = -8$ **[B1]**

Evaluates $\sqrt[3]{r} = \sqrt[3]{-\frac{1}{8}} = -\frac{1}{2}$ **[B1]**

8 **[A1]**

[Back to Question] *[Total Marks: 4]*

14 **(a)** $d = \sqrt{(-8-4)^2 + (2--3)^2}$ or $\sqrt{144+25}$ **[M1]**

13 **[A1]**

(b) $17 = \sqrt{(3-k)^2 + (-7-8)^2}$ **[M1]**

$289 = (3-k)^2 + 225$ **[M1]**

$(3-k)^2 = 64$ or $k^2 - 6k - 55 = 0$ **[M1]**

$k = -5$ and $k = 11$ **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** Forms three equations: $a+b+c=4$, $4a+2b+c=13$, $9a+3b+c=28$ **[M1]**

Eliminates c correctly to obtain two linear equations: $3a+b=9$ and $5a+b=15$ **[M1]**

$a = 3$ **[A1]**

$b = 0$ **[A1]**

$c = 1$ **[A1]**

(b) $[3(k+1)^2+1] - [3k^2+1] = 57$ or $6k+3=57$ **[M1]**

9 **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** States A = Curved Surface Area + Base Area $= 2\pi rh + \pi r^2$ **[M1]**

Factorises πr out to obtain $\pi r(2h+r)$ **[A1]**

(b) **(i)** Substitutes $r=4$ and $h=6$: $A = \pi(4)(2(6)+4)$ **[M1]**

$A = 4\pi(12+4) = 64\pi$ **[A1]**

(ii) Rearranges volume formula: $h = \frac{V}{\pi r^2}$ **[M1]**

Substitutes h into area formula: $A = 2\pi r\left(\frac{V}{\pi r^2}\right) + \pi r^2$ **[M1]**

Simplifies first term: $\frac{2\pi rV}{\pi r^2} = \frac{2V}{r}$, resulting in $A = \frac{2V}{r} + \pi r^2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Substitute $x=-2$ and $y=4$: $P = \frac{2(-2)^2 - 3(4)}{-2-4}$ **[M1]**

$P = \frac{2(4)-12}{-6} = \frac{8-12}{-6}$ **[M1]**

$\frac{2}{3}$ (or equivalent) **[A1]**

(b) Multiply both sides by $(x-y)$: $P(x-y) = 2x^2 - 3y$ **[M1]**

Expand: $Px - Py = 2x^2 - 3y$ **[M1]**

Collect terms with y on one side: $3y - Py = 2x^2 - Px$ **[M1]**

Factorise and divide: $y(3-P) = 2x^2 - Px \implies y = \frac{2x^2 - Px}{3-P}$ (or equivalent) **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** **(i)** $\frac{6^2 - (-2)^2}{6(-2)}$ **[M1]**

$-\frac{8}{3}$ **[A1]**

(ii) $\frac{0.5^2 - 4^2}{0.5(4)}$ **[M1]**

$\frac{-15.75}{2}$ **[M1]**

$-\frac{63}{8}$ **[A1]**

(b) $\frac{x^2 - 2^2}{2x} = \frac{3}{2}$ **[M1]**

$2(x^2 - 4) = 6x$ (or $x^2 - 3x - 4 = 0$) **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 8]*

19 **(a)** $C + Rh$ or $C + hR$ **[B1]**

(b) M1 for $45 + 32.50 \times 3.5$ or $45 + 113.75$

158.75 **[A1]**

[Back to Question] *[Total Marks: 3]*

20 Substitutes values correctly: $\frac{\sqrt{12.5 - 3.84}}{(-1.2)^3}$ **[M1]**

Evaluates numerator to 2.94... or denominator to -1.728 **[M1]**

-1.70 (accept $-1.703...$ rounded strictly to 3 sig figs) **[A1]**

[Back to Question] *[Total Marks: 3]*

21 $\pi \times (4.5)^2 \times 8.2 + \frac{2}{3} \times \pi \times (4.5)^3$ **[M1]**

521.6... + 190.8... or 712.5... **[M1]**

713 **[A1]**

[Back to Question] *[Total Marks: 3]*

22 **(a)** $45 + 32h$ (or equivalent) **[B1]**

(b) 157 **[B1]**

(c) Correct substitution: $(-3)^3 - 4(-3)(2.5)$ or $-27 - (-30)$ **[M1]**

3 **[A1]**

[Back to Question] *[Total Marks: 4]*

23 **(a)** $V = \frac{1}{3}\pi(6.4)^2(12.6)$ **[M1]**

540.5 **[A1]**

(b) $800 = \frac{1}{3}\pi r^2(14)$ **[M1]**

$r^2 = \frac{2400}{14\pi}$ or $54.567\ldots$ **[M1]**

7.39 **[A1]**

[Back to Question] *[Total Marks: 5]*

24 **(a)** $85(7) + 0.24(450)$ **[M1]**

703 **[A1]**

(b) $1475.60 = 85(14) + 0.24m$ **[M1]**

$1475.60 - 1190 = 0.24m$ (or $285.60 = 0.24m$) **[M1]**

1190 **[A1]**

[Back to Question] *[Total Marks: 5]*

25 **(a)** Substitute $h = 3.5$: $C = 45 + 35(3.5)$ **[M1]**

167.5 **[A1]**

(b) Substitute $C = 325$: $325 = 45 + 35h$ **[M1]**

$280 = 35h$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 5]*

26 **(a)** $A = \pi \times 5 \times \left(5 + \sqrt{5^2 + 12^2}\right)$ or 90π seen **[M1]**

283 (282.74...) **[A1]**

(b) $96\pi = \pi(6)\left(6 + \sqrt{6^2 + h^2}\right)$ **[M1]**

$16 = 6 + \sqrt{36 + h^2}$ or $\sqrt{36 + h^2} = 10$ **[M1]**

$36 + h^2 = 100$ or $h^2 = 64$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 6]*

27 **(a)** Substitutes $h = 4.2$ and $a = 6.5$ into formula: $V = \frac{1}{6}\pi(4.2)(3(6.5)^2 + 4.2^2)$ **[M1]**

$V = 317.53... = 317.5$ **[A1]**

(b) Multiplies by 6 and divides by πh: $\frac{6V}{\pi h} = 3a^2 + h^2$ **[M1]**

Subtracts h^2: $\frac{6V}{\pi h} - h^2 = 3a^2$ **[M1]**

$a^2 = \frac{2V}{\pi h} - \frac{h^2}{3}$ or $a^2 = \frac{1}{3}\left(\frac{6V}{\pi h} - h^2\right)$ **[A1]**

(c) $a = \sqrt{\frac{2(150)}{\pi(3)} - \frac{3^2}{3}} = \sqrt{31.83... - 3} = \sqrt{28.83...} = 5.37$ **[B1]**

[Back to Question] *[Total Marks: 6]*

28 Numerator correctly evaluated: $-4.5 - (-3.2)^2 = -4.5 - 10.24 = -14.74$ **[B1]**

Denominator correctly evaluated: $1.8 - \sqrt{0.64} = 1.8 - 0.8 = 1$ **[B1]**

-14.74 **[B1]**

[Back to Question] *[Total Marks: 3]*

29 **(a)** **(a)** $5.2(4.5) + 0.5(-9.8)(4.5)^2$ **[M1]**

-75.825 **[A1]**

(b) $-9.8 = 4.9t + \frac{1}{2}(-9.8)t^2$ **[M1]**

$4.9t^2 - 4.9t - 9.8 = 0$ or $t^2 - t - 2 = 0$ **[M1]**

$t = 2$ (rejects -1 as time must be positive) **[A1]**

[Back to Question] *[Total Marks: 5]*

30 **(a)** M1 for correctly substituting the values into the formula: $\frac{1}{3}\pi(12.5)(4.2^2 + 4.2 \times 7.8 + 7.8^2)$

M1 for 111.24 or 463.5π or 1456.1...

1460 **[A1]**

(b) M1 for correctly substituting the values into the square root part: $\sqrt{(7.8 - 4.2)^2 + 12.5^2}$

M1 for $\sqrt{169.21}$ or 13.008...

736.9 **[A1]**

[Back to Question] *[Total Marks: 6]*